

Silver Plating + Faraday's Law

Lecture & Instructor Notes

Audience: instructors and teaching assistants

Purpose: Teaching script, deep-dive notes, and coaching guidance

Session hook: Estimate how many atoms were deposited

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

1. Session overview & plan
2. Lecture talking points
3. Instructor deep-dive notes
4. Preparation reminders

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

Session 3 — Silver Plating and Faraday's Law

Hook: Calculate how many silver atoms you deposit and how thick the film is.

Learning outcomes

- Review galvanic vs electrolytic ideas from Sessions 1–2
- Plate silver onto a metal object with controlled current and time
- Use $Q = It$, $n = Q/F$, and mass/thickness relations
- Complete a pre-lab Faraday prediction before plating
- Discuss assumptions (100% efficiency, uniform coverage, graphite inert anode)

Session plan (90 min)

Time	Activity	Reference
0–8 min	Review Days 1–2 + hook	lecture.md §1–2
8–18 min	Teach Ag^+ reduction + Faraday pipeline	lecture.md §3
18–28 min	Quiz check for understanding	lecture.md §4
28–45 min	Interactive pre-lab calculation (required)	lecture.md §5
45–55 min	Estimate area; set up cell (graphite anode, ammeter in series)	experiment.md
55–70 min	Timed plating; log current	experiment.md
70–85 min	Recalculate with real I_{avg} ; compare to prediction	experiment.md + lecture.md §6
85–90 min	Debrief efficiency; bridge to Session 4	lecture.md §7

Key equations

$$\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow \text{Ag}(\text{s})$$

$$Q = I \times t$$

$$n_{\text{e}} = Q / F \quad (F = 96,485 \text{ C/mol})$$

$$n_{\text{Ag}} = n_{\text{e}}$$

$$m_{\text{Ag}} = n_{\text{Ag}} \times M_{\text{Ag}} \quad (M_{\text{Ag}} = 107.87 \text{ g/mol})$$

$$d = m_{\text{Ag}} / (\rho_{\text{Ag}} \times A) \quad (\rho_{\text{Ag}} = 10.49 \text{ g/cm}^3)$$

Files

File	Purpose
lecture.md	Full teaching script: review, plan, explanations, quiz, pre-lab
experiment.md	Bench procedure and data sheets

File	Purpose
instructor.md	Deep-dive coaching notes
materials.md / preparation.md	Logistics

Status

- ☐ AgNO₃ solution prepared (0.01 M, distilled water)
- ☐ Pre-lab worksheets printed (lecture.md §5)
- ☐ Graphite anodes staged
- ☐ Pilot plating run completed (graphite anode)
- ☐ Waste container labeled for silver

Session 3 — Lecture Notes: Silver Plating & Faraday's Law

Audience: instructors / TAs — use this as your teaching script.

Students see the shorter talking points during class; they do the worksheet in the interactive section before plating.

Deep-dive chemistry, efficiency, and coaching traps: [instructor.md](#).

Bench procedure: [experiment.md](#).

1. Summary of concepts from prior days

Spend **5–8 minutes** refreshing these ideas. Students need them today.

Session 1 — Fruit battery (galvanic cell)

Idea	What to remind them
Oxidation / reduction	Oxidation = lose e^- ; reduction = gain e^- (OIL RIG)
Anode / cathode (galvanic)	Anode = oxidation = negative (-) terminal; cathode = reduction = positive (+)
Electron vs ion path	Electrons travel in the wire ; ions travel in the juice
Series cells	Voltages add; current is still limited by the weak fruit electrolyte
Spontaneous chemistry	The reaction makes electricity — no battery required

One-sentence bridge: “Day 1: chemistry pushed electrons through a wire.”

Session 2 — Copper electroplating (electrolytic cell)

Idea	What to remind them
Electrolysis	Electricity forces a non-spontaneous reaction
Anode / cathode (electrolytic)	Signs flip relative to a battery: object to plate is on battery (-) = cathode = reduction
$\text{Cu}^{2+} \rightarrow \text{Cu}$	Copper ions in solution gain electrons and become solid metal on the object
Copper anode	Metal anode dissolved and replenished Cu^{2+} in the bath
Current & quality	Higher current → faster plating, but can make powdery / burnt deposits
Surface prep	Dirty / oily surfaces plate poorly

One-sentence bridge: “Day 2: electricity pushed copper atoms onto a nail — but we mostly watched, we didn’t count.”

What is new today (Session 3)

- Same **electrolytic plating** idea as Day 2
- Metal is **silver** ($\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$) — simpler math: **1 electron per silver atom**
- Anode is **graphite (inert)** — does **not** replenish Ag^+ (contrast with copper anode)
- Students **measure current and time**, then **calculate** moles, mass, atoms, and thickness with **Faraday's law**

One-sentence goal: *"Today we count how much silver we deposit using electricity as a measuring stick."*

2. Class plan (90 min)

Time	Block	What you do	Student product
0–8 min	Prior-days review + hook	Use §1; show a dull copper object	Oral answers to 2–3 review questions
8–18 min	Teach today's concepts	Use §3 explanations; board flowchart	Notes / vocab
18–28 min	Quiz check	Run §4 questions (cold call or mini-whiteboard)	Quick oral or written answers
28–45 min	Interactive pre-lab	§5 activity + calculation worksheet	Completed pre-lab sheet (required before plating)
45–55 min	Setup	Gloves/goggles; area estimate; wire cell with graphite anode + ammeter in series	Setup table filled
55–70 min	Plate	Timed run; log current every 30–60 s	Current log
70–85 min	Post-lab calc	Use their I_{avg} and t in the same Faraday pipeline	Thickness + atom estimate
85–90 min	Debrief + bridge	Efficiency, graphite vs Cu anode, preview Session 4	Exit ticket (optional)

Pacing tip: Do not start plating until the pre-lab calculation (§5) is checked. Groups that skip it will stall later and wire the ammeter wrong.

3. Detailed explanations to give students

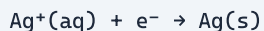
Say these ideas in your own words; keep the board organized.

3.1 Hook (say this)

"Yesterday we made copper appear on a nail. Today we plate silver — and I want more than 'it looks shiny.' If we run 0.02 amps for 5 minutes, can we calculate how many silver atoms landed — without weighing the coin?"

Answer: **Yes**, if we know how charge relates to electrons, and how electrons relate to silver atoms. That relationship is **Faraday's law**.

3.2 The silver half-reaction (board)



Explain:

- Ag^+ is a silver ion dissolved in the bath (from **0.01 M AgNO_3** , made with **distilled water** — tap water's chloride can make white AgCl and spoil the bath).
- One Ag^+ ion needs **exactly one electron** to become one silver atom.
- So: **moles of Ag deposited = moles of electrons** that went into plating silver (if every electron does that job).

Contrast with Day 2 (important):



Ask: "If the same number of electrons flows, do you deposit more moles of Ag or of Cu?"

→ More moles of **Ag** (because each Ag only needs one e^-).

3.3 What current and time mean

Symbol	Meaning	Units
I	Current = how fast charge flows	Ampere (A) = coulomb per second
t	Time current flows	Seconds (s)
Q	Total charge that passed	Coulomb (C)

$$Q = I \times t$$

Trap to say out loud: Multimeters often show **mA**.

20 mA = 0.020 A. If you forget to convert, your answer is off by **1000×**.

3.4 Faraday's constant

$$F = 96,485 \text{ C/mol e}^- \approx 9.65 \times 10^4 \text{ C per mole of electrons}$$

Explain: One mole of electrons (Avogadro's number of electrons) carries about 96,485 coulombs.

$$n_{\text{e}} = Q / F$$

Then for silver:

$$n_{\text{Ag}} = n_{\text{e}} \quad (\text{because } z = 1)$$

$$m_{\text{Ag}} = n_{\text{Ag}} \times 107.87 \text{ g/mol}$$

3.5 From mass to thickness (why we need area)

Silver has density $\rho = 10.49 \text{ g/cm}^3$.

$$\text{Volume } V = m / \rho$$

$$\text{Thickness } d = V / A = m / (\rho \times A)$$

A = plated surface area in cm^2 (estimate from coin/washer geometry).

Say this clearly: Calculated thickness will be tiny — often **less than one micrometer**. You will not measure it with a ruler. The calculation predicts an invisible-thin film; your eyes only see that it looks more silvery.

3.6 The full pipeline (draw as a flowchart)

$I \text{ (A)}, t \text{ (s)}$

$$\rightarrow Q = I \times t \text{ (coulombs)}$$

$$\rightarrow n_e = Q / F$$

$$\rightarrow n_{\text{Ag}} = n_e$$

$$\rightarrow m_{\text{Ag}} = n_{\text{Ag}} \times 107.87$$

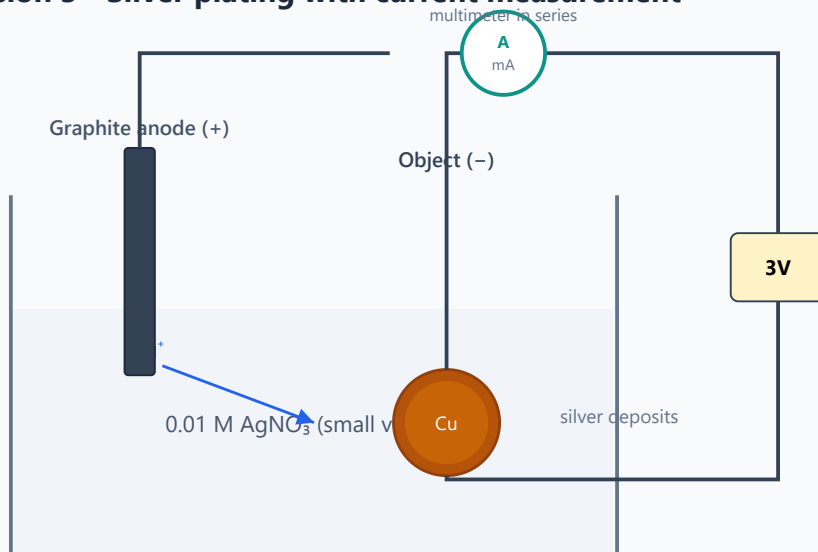
$$\rightarrow d = m_{\text{Ag}} / (\rho \times A)$$

Optional extra step after n_{Ag} :

$$N_{\text{atoms}} = n_{\text{Ag}} \times N_A \quad (N_A = 6.022 \times 10^{23} \text{ mol}^{-1})$$

3.7 Cell hardware today (say while showing the figure)

Session 3 - Silver plating with current measurement



$$Q = I \times t \rightarrow n = Q/F \rightarrow m_{\text{Ag}} \rightarrow \text{thickness } d = m/(\rho A)$$

Estimate plated area A before starting (coin face, washer ring, etc.)

Part	Connection / role
Object to plate	Battery (–) = cathode = Ag^+ reduced $\rightarrow$ Ag metal
Graphite rod	Battery (+) = anode = inert; does not dissolve to make Ag^+
AgNO_3 solution	Source of Ag^+ ions
Multimeter	On DC mA , wired in series (charge must go through the meter)

Contrast with Day 2: Copper anode replenished Cu^{2+} . Graphite does **not** replenish Ag^+ . The bath has enough Ag^+ for a short classroom run; Ag^+ slowly depletes. Faraday math for the **cathode deposit** still works.

3.8 Ideal vs real (set expectations before they plate)

Faraday thickness assumes:

1. Every electron plates silver (100% current efficiency)
2. Deposit is uniform
3. Your area estimate is correct
4. You used average current if I drifted

In real life, some current may make H_2 or other side reactions; edges plate faster; area is approximate. So: **the calculation is the ideal maximum / estimate**, then we compare to what we see.

3.9 Safety (brief but firm)

- Gloves and goggles **before** pouring AgNO_3
- AgNO_3 stains skin and clothes dark brown/black — permanent looking
- Small volumes only (20–30 mL)
- All silver waste $\rightarrow$ labeled container — **not** down the drain

4. Quiz questions (check understanding before the lab)

Use as **cold call**, pair-share, or 3-minute written quiz. Suggested answers in *italics*.

A. Review from Days 1–2

Q1. In the fruit battery, which electrode is oxidized — zinc or copper?

Zinc ($\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$). Copper is where reduction happens.

Q2. In copper electroplating, the object to be plated is connected to which battery terminal? Why?

Negative (–). That makes it the cathode so Cu^{2+} can gain electrons and become Cu metal.

Q3. True or false: In an electrolytic cell, the anode is always negative.

False. In electrolysis the anode is connected to battery (+). (In a galvanic cell the anode is –.)

Q4. Why did Day 2 use a copper anode instead of a random metal?

Copper anode dissolves and replenishes Cu^{2+} in solution.

B. Today's concepts

Q5. Write the half-reaction for plating silver from Ag^+ . How many electrons per silver atom?

$\text{Ag}^+ + e^- \rightarrow \text{Ag}$; one electron.

Q6. If $Q = 9.65 \text{ C}$ of charge plates silver at 100% efficiency, about how many moles of Ag form? ($F \approx 96500 \text{ C/mol}$)

$n = Q/F \approx 9.65/96500 \approx 1.0 \times 10^{-4} \text{ mol}$.

Q7. A student measures 25 mA for 200 s. What is Q in coulombs?

$I = 0.025 \text{ A}$; $Q = 0.025 \times 200 = 5.0 \text{ C}$.

Q8. Why must the multimeter be in **series** on the current setting, not clipped across the battery like a voltmeter?

Current mode measures charge flow through the circuit; across the battery can short / give nonsense and can damage the meter.

Q9. We use a graphite anode today. Does graphite replace the Ag^+ used up at the cathode?

No. Graphite is inert. Ag^+ comes from the AgNO_3 bath and depletes.

Q10. Why distilled water for AgNO_3 , not tap?

Tap water often has Cl^- , which precipitates AgCl and can ruin the bath.

C. Stretch (optional)

Q11. Same Q and same efficiency: do you deposit a larger mass of Ag or of Cu? ($M_{\text{Ag}} \approx 108$, $M_{\text{Cu}} \approx 63.5$; remember z .)

$n_{\text{Ag}} = Q/F$, $n_{\text{Cu}} = Q/(2F)$. Mass Ag = $(Q/F) \times 108$; mass Cu = $(Q/(2F)) \times 63.5$. Ag mass is larger in this comparison — good discussion.

5. Interactive activity + pre-lab calculation (required before plating)

Activity name: “Predict before you plate”

Format: Pairs or groups of 3–4. **15–20 minutes.**

Goal: Every group finishes one full Faraday calculation with **made-up but realistic** numbers before they touch AgNO_3 . Then they repeat the same steps with **real** I and t after plating.

Part 1 — Station cards (2 min)

Give each group a card (or write on board) with:

Parameter	Value
Planned current I	0.020 A (20 mA)
Planned time t	300 s (5 min)
Object	Copper coin, one face plates
Diameter	2.0 cm

Part 2 — Interactive prompts (do these live)

1. Area challenge (3 min):

Each group calculates ($A = \pi (d/2)^2$).

Cold call: “What A did you get?”

Expected: ($A = \pi(1.0)^2 \approx 3.14 \text{ cm}^2$).

If someone says 12.6, they used diameter as radius — fix it now.

2. Unit police (2 min):

Hold up a meter showing “20” on mA. Ask: “What number goes into $Q = It$?”

Chorus: **0.020 A**.

3. Human flowchart (3 min):

Five volunteers hold signs: $I, t \rightarrow Q \rightarrow n_e \rightarrow m_{Ag} \rightarrow d$.

Class shouts the formula linking each pair. Keeps energy up before math.

Part 3 — Calculation worksheet (8–10 min)

Students complete this table **before** getting silver solution. Check one line per group (initials) before they may start plating.

Constants: $F = 96,485 \text{ C/mol}$; $M_{Ag} = 107.87 \text{ g/mol}$; $\rho_{Ag} = 10.49 \text{ g/cm}^3$

Step	Formula	Work	Answer + unit
1. Area A	$\pi(r)^2$		___ cm^2
2. Charge Q	$I \times t$		___ C
3. Moles e^-	Q / F		___ mol
4. Moles Ag	$n_{Ag} = n_e$		___ mol
5. Mass Ag	$n \times 107.87$		___ g
6. Volume Ag	$m / 10.49$		___ cm^3
7. Thickness d	V / A		___ cm and ___ μm
8. (Optional) Atoms	$n \times 6.022 \times 10^{23}$		___ atoms

Target answers for the card above (for you):

Step	Result
A	$\approx 3.14 \text{ cm}^2$
Q	$0.020 \times 300 = \mathbf{6.0 \text{ C}}$
n_e	$6.0 / 96485 \approx \mathbf{6.22 \times 10^{-5} \text{ mol}}$
m	$\approx \mathbf{0.00671 \text{ g}}$
V	$\approx \mathbf{6.40 \times 10^{-4} \text{ cm}^3}$
d	$\approx \mathbf{2.04 \times 10^{-4} \text{ cm} \approx 2.0 \mu\text{m}}$

Step	Result
Atoms	$\approx 3.7 \times 10^{19}$ atoms

Sense-check to say: *"Two micrometers is about 1/50 the thickness of a sheet of paper. You'll see a color change, not a measurable slab."*

Part 4 — Prediction statement (1 min)

Each group writes one sentence:

"If we plate at about 20 mA for 5 minutes on area $\approx$ ___ cm², we expect roughly ___ μ m of silver (ideal Faraday)."

After the real run, they recalculate with **I_{avg}** and actual t and compare to this prediction.

Exit check before chemicals

Group may start plating only when:

- ☐ Pre-lab table complete
- ☐ Instructor/TA spotted Q and A
- ☐ Gloves + goggles on
- ☐ They can point to: cathode object (–), graphite anode (+), ammeter in series

6. During / after the experiment — what to emphasize

1. Log current often → use **I_{avg}**, not only the first reading.
2. Same Faraday pipeline as the pre-lab — only the numbers change.
3. If calculated d seems absurd (cm thick or 10^{–12} cm): check mA→A and area.
4. Debrief questions:
5. Did the coin look more silvery?
6. Was real I close to 20 mA?
7. Why might real thickness be less than ideal? (efficiency, area error)
8. How is graphite anode different from Day 2 copper anode?

7. Bridge to Session 4

"We've been counting metal atoms with electric charge. Next session we split water into hydrogen and oxygen — first with plain tap water (slow), then with sodium sulfate (fast) — measure current for rate, then collect gases to see the 2:1 molecule ratio."

8. Vocabulary checklist

- ☐ Electrolytic cell vs galvanic cell

- [] Cathode / anode (electrolytic signs)
 - [] Faraday constant
 - [] Coulomb, ampere
 - [] Current efficiency (intro)
 - [] Inert anode (graphite) vs sacrificial / metal anode
-

9. Materials you need in hand while teaching

- Dull copper coin/washer for hook
- Board markers; flowchart space
- Pre-lab cards or projected table
- Printed worksheet (§5 Part 3) — one per student or group
- Demo ammeter showing mA
- Graphite rod + object to show polarity with battery pack (dry demo OK)

Session 3 — Instructor Notes: Silver Plating + Faraday Deep Dive

Audience: instructors and TAs only.

Classroom teaching script (review, quiz, pre-lab activity): lecture.md.

Bench procedure: experiment.md.

Why this session exists

Students saw copper move in Session 2. Now they **quantify** electrodeposition: charge → moles of electrons → moles of metal → mass → thickness. Silver is ideal because $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$ is 1:1 (simpler stoichiometry than $\text{Cu}^{2+} + 2\text{e}^-$). This is the week's heaviest math session — go slowly on units.

Atomic / electronic foundations for silver

Silver ion and electron count



- Silver atom (Ag, $Z = 47$) loses one valence electron → Ag^+ .
- Plating reverses that: one electron returns → one Ag atom deposited.
- Therefore: **$n_{\text{Ag}} = n_{\text{e}}$** (moles silver = moles electrons) when plating is 100% efficient.

Compare with copper from Session 2:



Mention this contrast so students see why the electron count in the half-reaction matters.

Why dilute AgNO_3

Silver nitrate provides Ag^+ . Classroom concentration **0.01 M** keeps:

- Cost down
- Staining / hazard manageable
- Thin films that still look metallic

Safety (non-negotiable): AgNO_3 stains skin/clothes permanently dark; gloves + goggles; collect all silver waste.

Faraday's law — instructor mastery

Charge from current and time

$$Q = I \times t$$

Symbol	Unit	Meaning
I	ampere (A) = C/s	Current
t	second (s)	Time
Q	coulomb (C)	Total charge passed

Trap: students record mA but forget to convert. 20 mA = 0.020 A.

Faraday constant

$$F = 96,485 \text{ C/mol e}^- \approx 9.65 \times 10^4 \text{ C/mol}$$

Meaning: one mole of electrons carries ~96,485 C.

$$n_{\text{e}} = Q / F$$

Stoichiometry factor (general form)

For a reduction requiring **z** electrons per metal ion:

$$n_{\text{metal}} = n_{\text{e}} / z = Q / (z F)$$

Metal ion	z	Example
Ag ⁺	1	Session 3
Cu ²⁺	2	Session 2
Al ³⁺	3	not used here

Mass and thickness

$$m = n \times M$$

$$V = m / \rho$$

$$d = V / A$$

Quantity	Silver value
Molar mass M	107.87 g/mol
Density ρ	10.49 g/cm ³
A	geometric plated area (cm ²) — biggest student error source

Worked example (memorize this board version)

Given: I = 0.020 A, t = 300 s, A = 10 cm², Ag plating, z = 1.

Step	Math	Result
Q	0.020×300	6.0 C
n _e	$6.0 / 96485$	6.22×10^{-5} mol
n _{Ag}	= n _e	6.22×10^{-5} mol
m	$6.22 \times 10^{-5} \times 107.87$	6.71×10^{-3} g
V	$m / 10.49$	6.40×10^{-4} cm ³
d	$V / 10$	6.4×10^{-5} cm = 0.64 μm

Sense-check: visible silver films are often tenths of a micrometer to a few micrometers — students' numbers should be in that ballpark if I, t, A are sane.

Average current when I drifts

If students log I every 30–60 s:

$$I_{\text{avg}} \approx (I_0 + I_1 + \dots + I_n) / (n+1)$$

or trapezoidal estimate. Then $Q \approx I_{\text{avg}} \times t_{\text{total}}$.

Do **not** use only the first reading if current falls (common as concentration polarization / surface changes).

Ideal vs real plating (current efficiency)

Faraday assumes **every** electron reduces Ag⁺. Reality:

$$\text{current efficiency } \eta = (\text{charge used for Ag plating}) / (\text{total charge}) \leq 1$$

Loss channels:

- H₂ evolution
- Impurity reductions
- Non-uniform current (edges plate more — “dog-boning”)
- Flaky deposit that rinses off

If calculated thickness looks “too thick” vs visual: area underestimate or optimistic $\eta = 1$. If “too thin”: efficiency < 1, area overestimate, or current logged wrong.

Electrochemical series note for Ag

Couple	E° (V)
Ag ⁺ /Ag	+0.80
Cu ²⁺ /Cu	+0.34

Couple	E° (V)
H ⁺ /H ₂	0.00

Ag⁺ is a strong oxidant relative to H⁺ and Cu²⁺ — silver plates readily. That is why even dilute AgNO₃ works visually. It is also why silver ions are “eager” to reduce and why waste must be handled carefully (and why Ag stains via reduction on skin/organics).

Anode choice:

- **Planned anode: graphite (inert).** Ag⁺ comes only from the bath and depletes during the run; O₂ evolution at the anode is possible. Faraday calculations for **cathode** Ag deposit remain valid for short classroom runs if you start with enough 0.01 M AgNO₃ (20–30 mL).
- **Contrast with Session 2:** copper plating used a **copper** anode that replenished Cu²⁺. Here we deliberately use graphite — cheaper and we do not need silver metal anodes. Mention this difference so students do not assume every plating bath needs a metal anode of the same element.

Experiment coaching

Area estimation — coach aggressively

Object	Reasonable approach
Coin face	$A = \pi(r)^2$ per face; $\times 2$ if both faces plate
Washer	$\pi(R^2 - r^2)$
Irregular	Length $\times$ width of immersed region; say so

Students who invent a huge area get absurdly small thickness and think Faraday “failed.”

Ammeter in series

Multimeter must be **in series** on DC current range — not across the cell like a voltmeter (that shorts / misreads). Walk each station once.

Timing discipline

Start stopwatch when current is flowing and stable. Stop when disconnected. Gaps in logging → poor I_{avg}.

What success looks like

- Whitish / silvery sheen on copper object
- Not necessarily mirror-bright (classroom baths lack brighteners)
- Calculation completed with units at every step

Misconceptions

Misconception	Correction
" $Q = I + t$ "	$Q = I \times t$
"Use mA directly in $Q = It$ "	Convert to amperes
"Cu and Ag use the same z "	Ag^+ needs $1e^-$; Cu^{2+} needs $2e^-$
"Thickness must match a ruler"	Micrometers are invisible to a ruler; appearance is qualitative
"Faraday is wrong if deposit looks thin"	Efficiency and area errors dominate

Board plan

1. Write $\text{Ag}^+ + e^- \rightarrow \text{Ag}$; circle "1 electron per atom."
2. Flowchart: $I, t \rightarrow Q \rightarrow n_e \rightarrow n_{\text{Ag}} \rightarrow m \rightarrow V \rightarrow d$.
3. Full numerical example with units.
4. Discuss why I_{avg} beats a single reading.
5. Bridge to Session 4: next we count **gas molecules** by volume ratio, not metal atoms by charge.

Optional enrichment

- Derive z from half-reaction for Cu plating retroactively (Session 2 data).
- Mention electrolysis mass production (Al from Hall–Héroult — huge Faraday consumer).
- Introduce faradaic efficiency formally if a student is ready.

Session 3 — Preparation

Before class (instructor)

- ☐ Prepare 0.01 M AgNO_3 with **distilled/deionized water** (not tap); volume for all groups + 20% extra
- ☐ Pilot plate one object: record I_{avg} , t , appearance
- ☐ Verify 3 V packs deliver stable $\sim 10\text{--}30$ mA with a **graphite anode**
- ☐ Stage clean graphite anodes (one per station)
- ☐ Print pre-lab Faraday worksheets (lecture.md §5) — one per group
- ☐ Prep “station cards” with I , t , diameter for the interactive activity
- ☐ Confirm silver waste disposal with school/facility protocol

AgNO_3 handling

- Wear gloves when preparing
- Work in ventilated area
- Label all bottles clearly
- Store away from organic materials and light if possible

Day-of setup

- ☐ Small volumes pre-measured per station (reduces spills)
- ☐ Goggles enforced at door
- ☐ Waste container visible and labeled
- ☐ Demo calculation on board with example numbers (before students start)

Timing note

Calculations take longer than expected for some groups. Have I_{avg} pre-computed on board as check, or provide a spreadsheet template.

Open questions / notes

- Stock AgNO_3 concentration on hand:
- Volume prepared (mL):
- Pilot I_{avg} (mA):
- Pilot plating time (s):